
Plan Overview

A Data Management Plan created using DMPonline

Title: Graffiti Across Vienna

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Affiliation: Linnaeus University

Funder: Swedish Research Council

Template: Swedish Research Council Template

Project abstract:

Graffiti Across Vienna is a Unity 3D game that lets users experience a showcase of graffiti from all districts of Vienna by sitting in a tube and viewing them from the window. Using digitised images of graffiti captured by yours truly, the project documents and presents the city's diverse street art in a playful, immersive format. By bringing these photographs into a navigable 3D environment, the project aims to let users explore Vienna's graffiti culture remotely while also preserving a visual record of artworks that are often temporary.

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Graffiti Across Vienna

General Information

Graffiti Across Vienna

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Version 1

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Description of data - reuse of existing data and/or production of new data

All images used in the project will be collected by the project creator using a smartphone. These images will first be stored in Google Cloud, where they will be processed and prepared for use. Once the processed versions are created, any raw files that are no longer required will be deleted. The processed images will then be uploaded to Omeka, where they will be stored indefinitely as part of the project's dataset. The same processed images will also be reused within the Unity game environment. No external datasets will be reused; all data is original to the project.

The primary data type will be images, collected as digital photographs and stored in JPEG or PNG format. These will form the core dataset. Alongside the images, a small amount of metadata will be created in Omeka (for example: title, district, date, and description) and stored in standard text-based formats, such as Dublin Core fields. The total volume of data is expected to remain relatively small, likely between 200 and 500 MB, depending on the number of images collected and their resolution. No additional external data sources will be used.

Documentation and data quality

The material will be documented using a clear file name and title for each image (e.g., "smiling_cat"). Additional contextual information—such as district, capture date, and a short description—will be recorded in Omeka to ensure each item is easy to identify and understand. All metadata will follow the Dublin Core standard, as supported by Omeka, using fields such as Title, Date, Description, Coverage (location), and Rights. The collection method (smartphone photography in public space) will also be noted to provide basic context for how the material was created. Images reused in the Unity project will rely on the same metadata created in Omeka. Unity assets will be organised in a simple, consistent folder structure, but no separate metadata schema will be created for Unity, as the images will retain their documentation from Omeka.

Data quality will be safeguarded by keeping the collection and processing workflow simple and consistent. All images will be captured by the project creator using the same smartphone camera to ensure uniform quality. Each image will be checked immediately after capture to confirm that it is clear, correctly framed, and free of identifiable individuals. During processing, filenames, titles, and metadata entries will be reviewed to ensure they are accurate and consistent. Any images that are blurry, incorrectly documented, or cannot be anonymised will be deleted. As the dataset will be small and manually curated, quality control will be maintained through direct inspection rather than automated validation.

Storage and backup

During the research process, all images will be stored in Google Cloud, which provides secure, cloud-based storage and automatic redundancy. This ensures the data remains safe even if the smartphone used for collection is lost or damaged. Processed images and their associated metadata will also be stored in Omeka, which maintains its own server-level backups and preserves the metadata structure. In addition, the Unity project files and processed images will be backed up through GitHub, providing version control and an additional layer of protection. Access to all storage locations will be restricted to the project creator.

Data security will be maintained by avoiding the collection of personal or sensitive data whenever possible. The project will follow GDPR-aligned practices by:

- Avoiding identifiable individuals during image capture
- Removing sensitive details through blurring or cropping when needed

- Deleting non-anonymisable images to ensure no personal data is retained

All images and metadata will be stored in secure, access-controlled environments (Google Cloud, Omeka, and GitHub). Access to these platforms will be restricted to the project creator through password-protected accounts and, where available, two-factor authentication. No data will be shared with third parties during the research process, and no raw files containing identifiable information will be retained. This will ensure that both the data and its associated metadata remain protected throughout the project.

Legal and ethical aspects

Data handling will follow relevant legal requirements by ensuring that no personal or sensitive data is collected or retained. The project will avoid capturing identifiable individuals, remove sensitive details through blurring or cropping when necessary, and delete any images that cannot be anonymised. This approach aligns with GDPR principles of data minimisation and privacy by design. Confidentiality will be maintained by storing all data in secure, access-controlled environments, with access restricted to the project creator. No data will be shared with third parties during the research process. Intellectual property rights will be safeguarded by ensuring that all images are created by the project creator and therefore do not infringe on external copyright. Metadata created in Omeka follows standard, non-copyrighted Dublin Core fields. Any outputs produced in Unity will reuse only the project's own images and will not incorporate third-party assets.

Ethical data handling will be ensured by designing the project to minimise risk and avoid collecting information that could identify or harm individuals. Images will be captured in public spaces with care taken to avoid photographing identifiable people. If individuals or sensitive details appear unintentionally, the images will be either anonymised through blurring or cropping, or deleted entirely. This approach aligns with ethical principles of respect, privacy, and minimisation. Only the project creator will have access to the data during the research process, ensuring that no material is shared or viewed unnecessarily. All data will be used solely for the project, and no images or metadata will be repurposed in a way that could misrepresent the subjects or context. The project will rely exclusively on original material created by the researcher, avoiding ethical issues related to third-party ownership or misuse of external content.

Accessibility and long-term storage

The primary research dataset—consisting of processed images and their associated Dublin Core metadata—will be made accessible through the project's Omeka site. Omeka will serve as the long-term public repository, providing stable access to both the images and their descriptive metadata. Data will be made available once processing and documentation are complete, with no embargo anticipated. Only processed, anonymised images will be shared. Raw image files will not be made accessible, as they are deleted after processing and may contain identifiable information. Metadata (e.g., title, district, date, description) will be openly available alongside the processed images. Unity project files may also be made publicly accessible at the conclusion of the project, depending on the game's final structure and whether any third-party assets are included. If released, these files would be shared through a public GitHub repository and would include only materials created by the project creator or licensed for open reuse. Any Unity files containing proprietary or restricted assets will remain private. Reuse of the dataset will be permitted for non-commercial, academic, or educational purposes with appropriate attribution. Ethical constraints apply: users may not attempt to re-identify individuals or reconstruct deleted raw data. No additional restrictions beyond these standard ethical and attribution requirements are expected.

Long-term storage of the research data will be safeguarded by the project's creator, who will be responsible for maintaining and overseeing all archived materials. The processed images and their associated Dublin Core metadata will be preserved in Omeka, which provides stable, structured, and durable storage suitable for long-term access. Omeka's server-level backups and preservation-oriented design further support the integrity and longevity of the dataset. Only data that is complete, anonymised, and relevant to the project's final outcomes will be selected for long-term storage. This includes the processed images and their metadata, as well as any Unity project files that the creator chooses to make publicly available. Raw image files will not be retained, as they are deleted after processing and may contain identifiable information that is not appropriate for long-term preservation. If Unity files are selected for long-term storage, they will be archived either in a public GitHub repository or in another suitable repository chosen by the project creator. Only files created by the researcher or licensed for open reuse will be included to ensure that long-term storage does not introduce copyright or licensing issues.

No specialised systems or proprietary software will be required to understand or reuse the data in the long term. The core dataset will consist of processed images in standard formats (JPEG/PNG) and metadata recorded using the Dublin Core schema, both of which can be accessed and interpreted using widely available tools. The images and metadata stored in Omeka will remain accessible through standard web browsers, and the metadata can be exported or read in common text-based formats. If Unity project files are made publicly available, they will be optional supplementary materials rather than a requirement for understanding or analysing the dataset. These files can be opened with the free Unity Editor, but they are not required to access the primary research data. Overall, the dataset will be designed to be platform-independent and usable without reliance on specialised or proprietary systems.

For this project, the use of unique, persistent identifiers such as Digital Object Identifiers (DOIs) is not necessary. As this is a master 's-level project with a relatively small dataset, the materials will be made accessible primarily through Omeka, which provides stable item URLs and structured metadata suitable for long-term reference. These persistent URLs will be sufficient for citation, access, and reuse within the project's expected scope. If any Unity project files are made publicly available, they may be hosted on a public GitHub repository, which also provides stable versioned links. However, assigning DOIs or other formal persistent identifiers is not required for the intended academic use and scale of the project. Should institutional requirements change in the future, or if the dataset is later deposited in a repository that supports DOI minting, persistent identifiers could be added at that stage. For the purposes of this project, however, they are not needed.

Responsibility and resources

All responsibilities for data management, both during the research project and after its completion, will fall to the project's creator. This includes the organisation, documentation, storage, backup, and quality control of all data and metadata throughout the project's duration. After the project ends, the project's creator will also remain responsible for ongoing management and long-term preservation of the selected data. This includes maintaining

the Omeka collection, overseeing any publicly released Unity files or GitHub repositories, and ensuring that all stored materials will continue to meet ethical, legal, and technical requirements over time.

Data management for this project will require minimal resources. All tasks related to storage, backup, documentation, and long-term preservation will be carried out by the project's creator as part of the normal research workflow. The primary tools used for data management will be Omeka, GitHub, and Unity:

- Omeka will provide long-term storage and public access for processed images and metadata.
- GitHub will support version control and backup for the Unity project files.
- Unity will be used to integrate the processed images into the game environment, but it will not introduce additional data-management costs. Unity's free version is sufficient for the project, and any Unity files made public will be stored via GitHub.

All three platforms can be used at no financial cost at the scale required for a master's project. To ensure the data meets the FAIR principles:

- Findability: Omeka provides stable URLs and structured Dublin Core metadata.
- Accessibility: Data will be accessible through standard web browsers and common file formats.
- Interoperability: JPEG/PNG images and text-based metadata ensure compatibility with widely used tools.
- Reusability: Clear documentation, consistent metadata, and the use of original, rights-free images support responsible reuse.

No additional software, paid services, or specialised infrastructure will be required to meet FAIR standards.